

AI TRUST, SAFETY & GOVERNANCE

GroundCheck / AuditLens

An LLM output verification and audit-trail layer — built on Exasol Personal

Exasol AI + Data Challenge 2026 · Track: AI Trust, Safety & Governance

Built by Raghul K and team

The problem

LLMs are shipping into production faster than anyone can govern them

1

No ground truth check

Model outputs go straight to users with nothing verifying them against the organization's own source documents.

2

No audit trail

When a model says something wrong, there's usually no record of what was checked, when, or why it passed.

3

No accountability across models

Teams swap models and providers constantly, with no consistent way to compare how often each one hallucinates.

4

Compliance exposure

Regulated industries (finance, healthcare, legal) need evidence of oversight — not just a chatbot that "seems fine."

Our solution

GroundCheck / AuditLens verifies every LLM response against real source documents — and logs the result

1

Verify

Every claim in a response is split out and matched against a knowledge base of trusted source documents.

2

Score

Each claim gets a groundedness score. Unsupported claims are flagged with the reason and closest source.

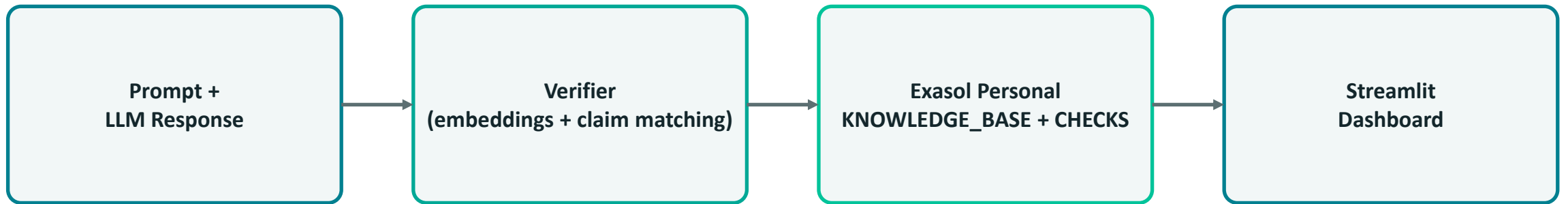
3

Log

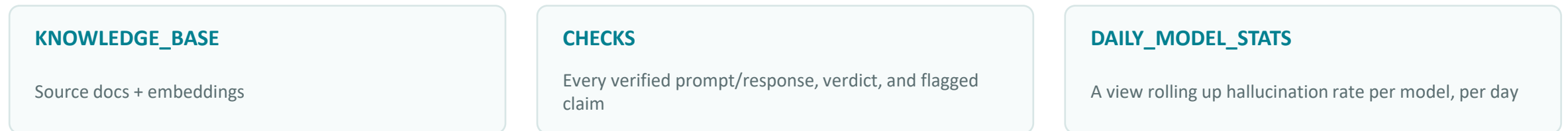
Prompt, response, verdict, confidence, and flagged spans are written to Exasol as a permanent, queryable audit row.

How it works

Exasol Personal sits at the center — as the knowledge store and the audit log



Exasol is queried live by the dashboard — no separate warehouse or cache in between



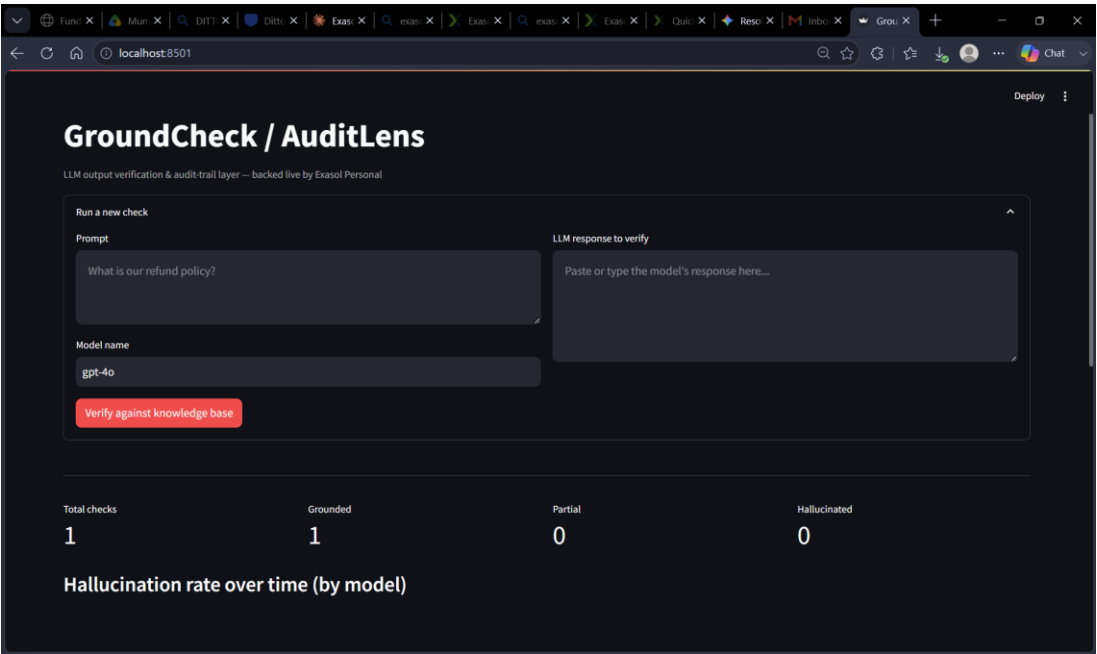
Exasol Personal, doing real work

Not a bolt-on data store — the system of record the whole project depends on

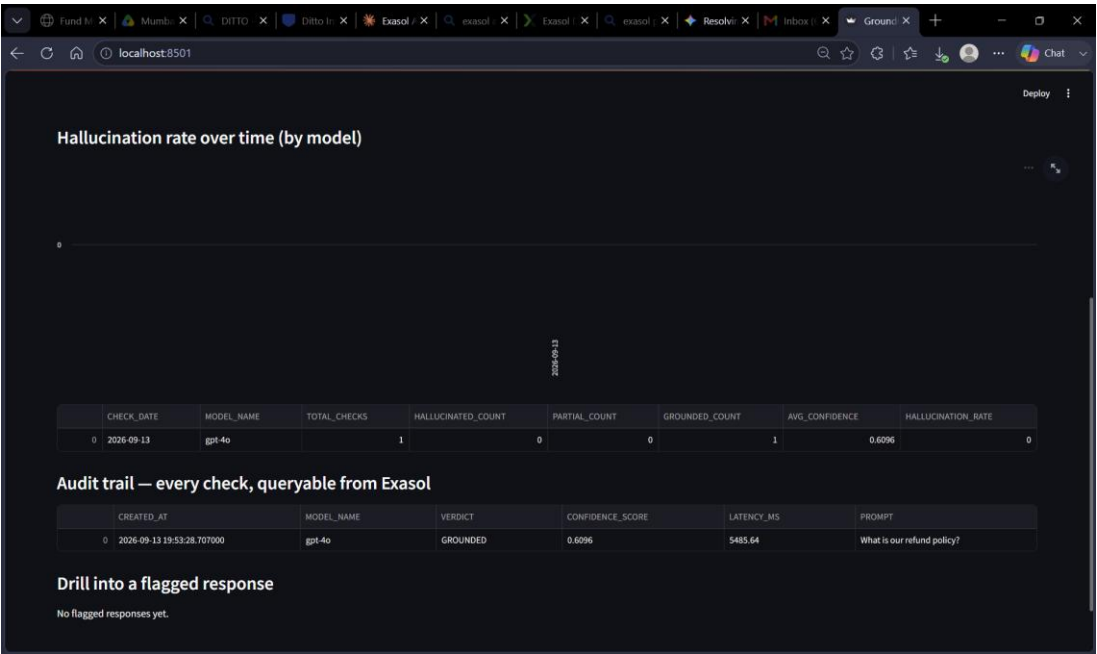
- Runs entirely local via Exasol Personal (starter kit) — full in-memory performance, no cloud dependency, private by default.
- KNOWLEDGE_BASE stores source-of-truth documents and their embeddings — the ground truth every claim is checked against.
- CHECKS is an append-only audit log: every prompt, response, verdict, confidence score, and flagged span, queryable by anyone with access.
- A native SQL view (DAILY_MODEL_STATS) rolls this up into hallucination rate per model per day — no external BI tool needed.
- The dashboard reads Exasol directly on every refresh, so what judges see on screen is a live query, not a cached export.

Live demo

Submit a response → get a verdict → see it logged and charted from Exasol



Prompt + response submitted; verified GROUNDED at a 0.61 confidence score.



Same check, now visible in Exasol's audit trail and the hallucination-rate view.

Impact & fit with the judging criteria

Criterion	Weight	How GroundCheck delivers
Innovation & problem impact	25%	Turns AI governance from a policy document into a running, queryable system.
Effective use of Exasol Personal	25%	Exasol is the knowledge base, the audit log, and the analytics layer — not decoration.
Technical excellence	20%	Embedding-based claim verification, a normalized audit schema, and a live-querying dashboard.
Solution design & UX	15%	One screen: submit, verify, see the verdict, drill into any flagged claim.
Presentation & demo	10%	A 3-minute run-through of the full loop, end to end, on real data.
GitHub & documentation	5%	Clean README, schema, .env template, and a step-by-step run guide.

Trust, but verify — automatically.

GroundCheck / AuditLens turns AI governance into a running system, not a policy binder — built entirely on Exasol Personal.

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